

Lecture D2: Slope Fields, Euler's method, Exist. & Unique.

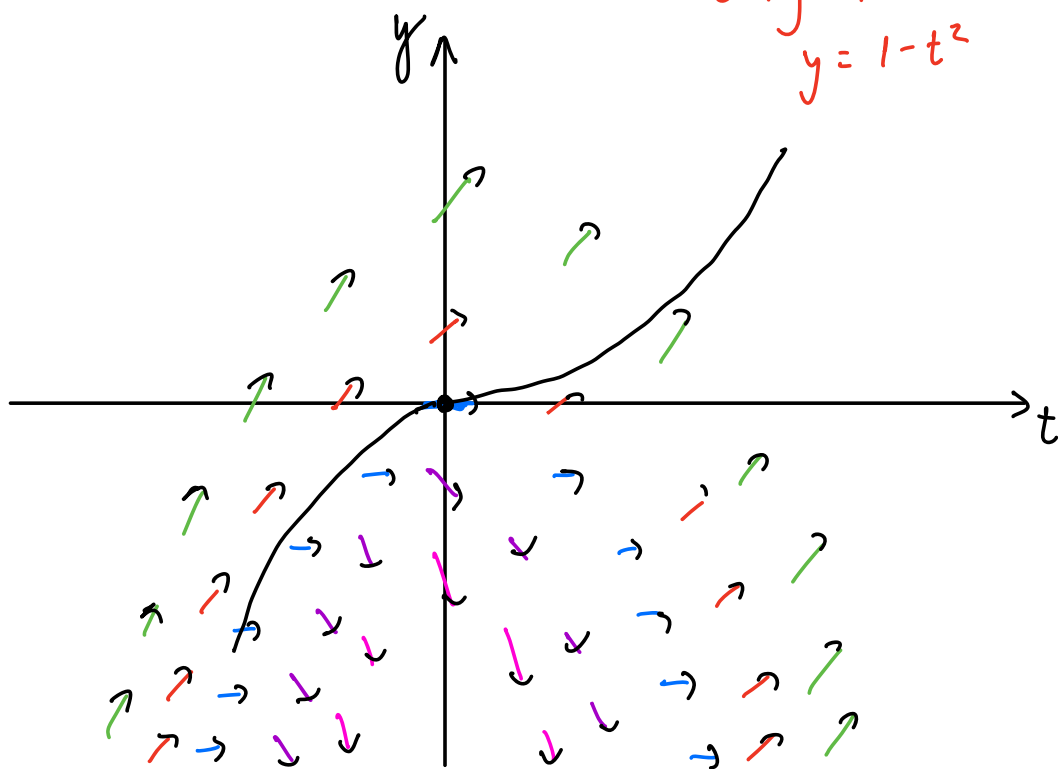
Slope Fields: graph with mini-tangent lines.

- Disclaimer: ~~we~~^I will use vector field and slope field interchangeably. the main difference is that one lacks sense of "direction".

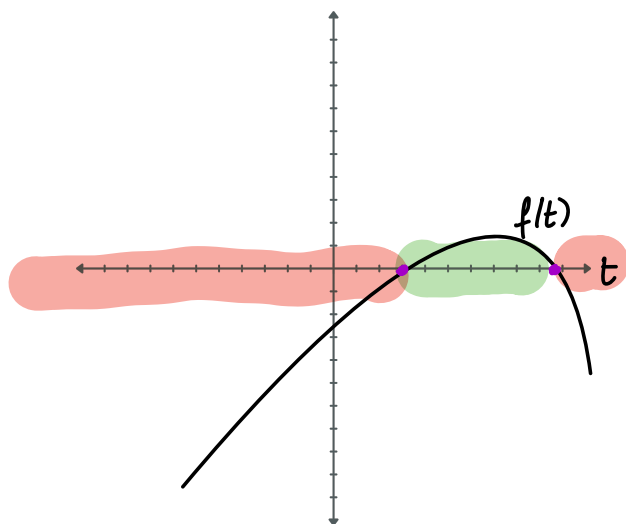
Example: $\frac{dy}{dt} = t^2 + y$ Draw slope field + solutions

$$t^2 + y = 0 \\ y = -t^2$$

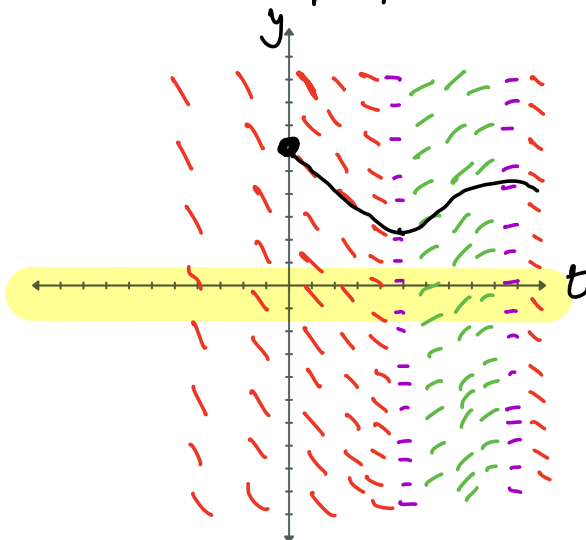
$$t^2 + y = 1 \\ y = 1 - t^2$$



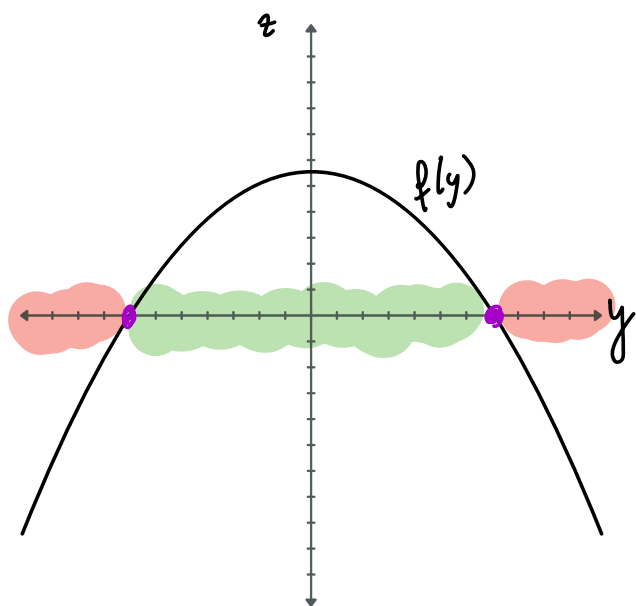
Suppose that $\frac{dy}{dt} = f(t)$ and the graph of f is



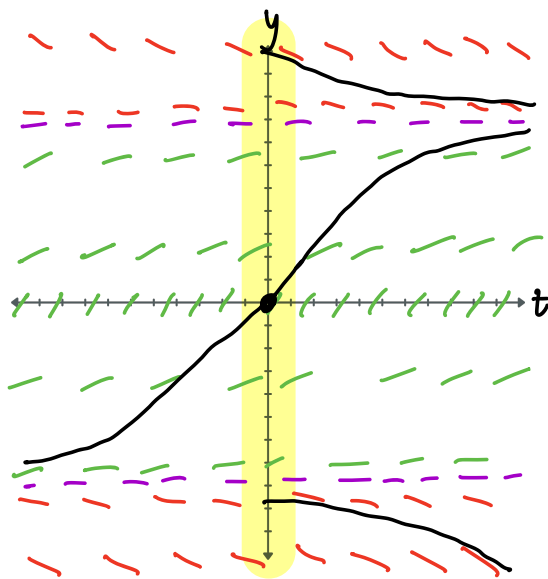
Sketch the slope field.



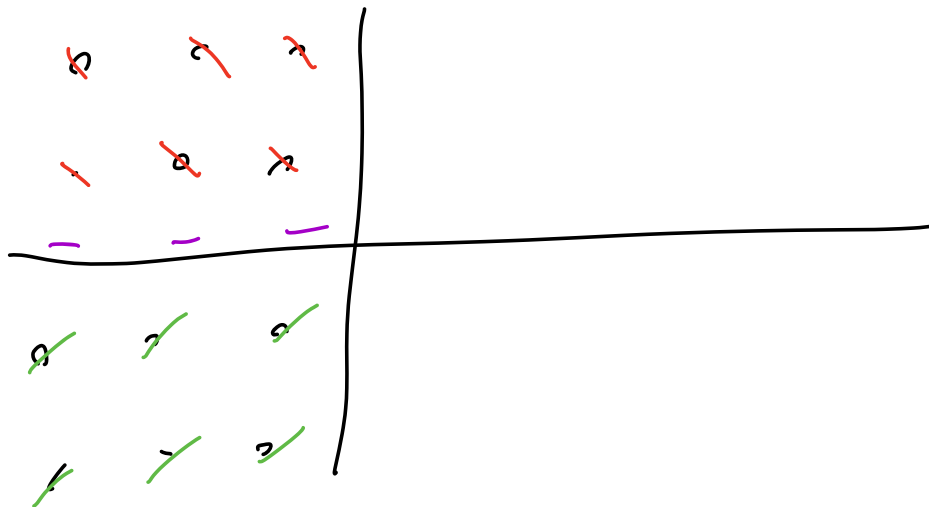
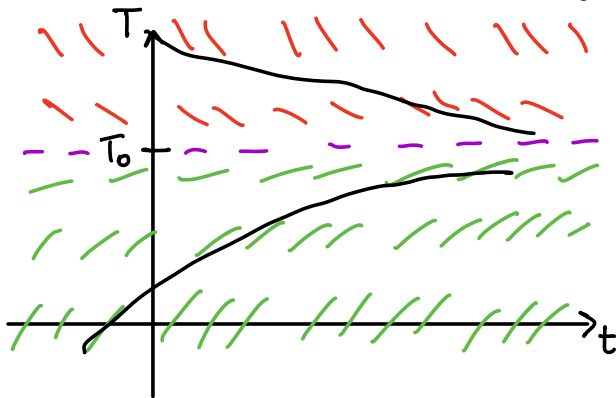
Suppose that $\frac{dy}{dt} = f(y)$ and the graph of f is



Sketch the slope field.



Ex. Newton's law of cooling: $\dot{T} = k(T_0 - T)$



EULER'S METHOD.

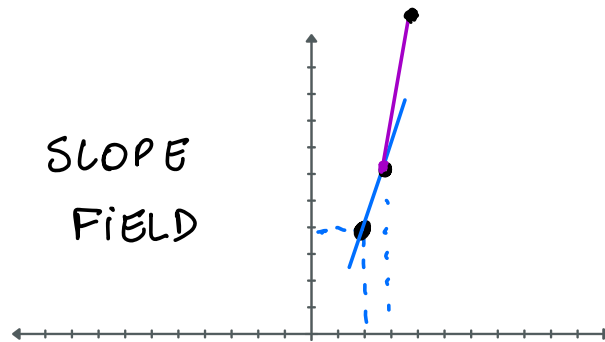
CONSIDER THE INITIAL VALUE PROBLEM

$$\frac{dy}{dt} = f(t, y) \quad , \quad y(t_0) = y_0$$

NUMERICALLY, WHAT COULD YOU DO?

SAY $\begin{cases} \frac{dy}{dt} = \frac{2y}{t} = f(t, y) \\ y(2) = 4 \end{cases}$

$$f(2, 4) = \frac{2 \cdot 4}{2} = 4$$



Estimate $y(3), y(4), y(5)$ using $\Delta t = 1$.

$$f(x+h) = f(x) + h \cdot f'(x)$$

$$y(2+1) = y(2) + 1 \cdot f(2, y(2)) = 4 + 1 \cdot f(2, 4) = 4 + 4 = 8$$

$$y(4) = y(3) + 1 \cdot f(3, y(3)) = 8 + 1 \cdot f(3, 8) = 8 + \frac{2 \cdot 8}{3}$$

$$y(t_k) = y(t_{k-1}) + (t_k - t_{k-1}) \cdot f(t_{k-1}, y(t_{k-1}))$$

$$y_k = y_{k-1} + \Delta t f(t_{k-1}, y_{k-1})$$

Estimate $y(3)$ using $\Delta t = 0.5$.

$$y' = \frac{2y}{t}$$

Stepsize: Δt

$$\Delta t = 1 \Rightarrow y(3) = 8.$$

$$\int \frac{1}{y} dy = \int \frac{2}{t} dt$$

"Homework": Estimate $y(3)$ using $\Delta t = 0.5$.

• Compare to the solution. $y(t) = t^2$.



Formulas

◦ Intuition:

◦ Taylor's series: $y(t)$ around $t=0$. $\frac{dy}{dt} = f(t, y)$

$$y(t) = y(0) + \underbrace{y'(0)}_{\uparrow} \cdot (t-0) + \frac{y''(0)}{2!} \cdot (t-0)^2 + \frac{y^{(3)}(0)}{3!} (t-0)^3 + \dots$$

$$y(t) = y_0 + f(0, y_0) \cdot t + \underbrace{o(t^2)}_{\rightarrow 0}$$

$$y(t) - y_0 - f(0, y_0)t = \underbrace{\frac{y''(0)}{2!}}_{\downarrow y'' = (y')' = (f)' = f'} t^2 + o(t^3)$$

$$\textcircled{+} \quad y(t) - y_0 - f(0, y_0)t = \frac{f'(0)}{2!} t^2 + \underset{\uparrow \text{ order}}{o(t^3)}$$

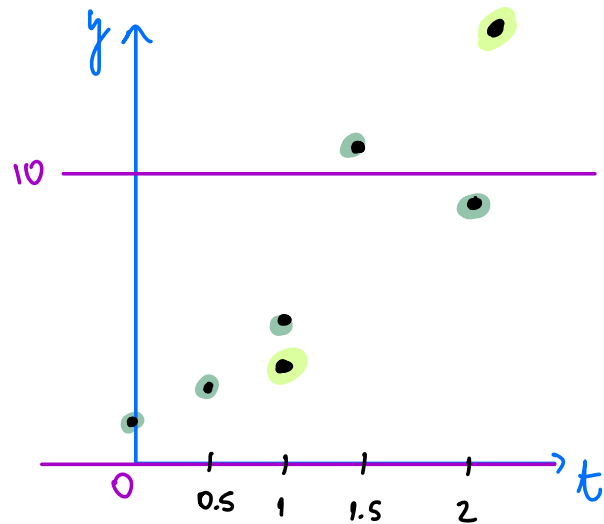
idea here was to showcase the relation with Taylor series, it is not in the book.

Ex: $y' = 3y(1 - \frac{y}{10})$, $y(0) = 2$.

Estimate $y(2)$ using

a) $\Delta t = 1$

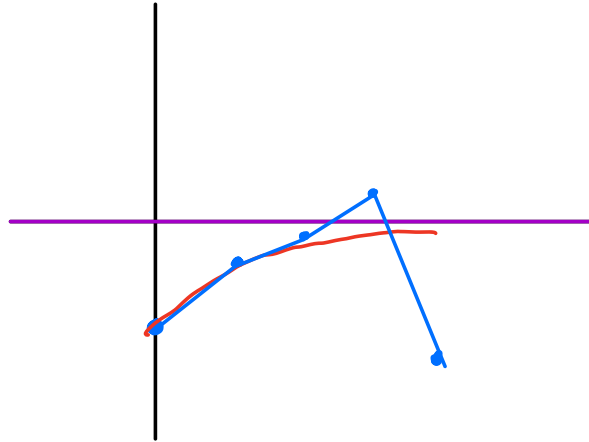
b) $\Delta t = 0.5$



good exercise for practice and see the error.

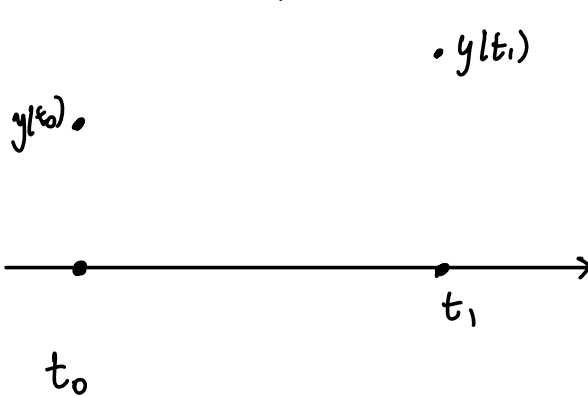
the problem with numerical results: *imprecision*

Ex:
$$\begin{cases} \frac{dy}{dt} = -e^t \cdot y \\ y(0) = -1 \end{cases}$$



(similar to figures 1.41 and 1.42 in page 61)

Error Estimate for 1-step.



$$\frac{dy}{dt} = f(t, y)$$

Find an upper bound
for f' : $|f'| \leq M$

$$y(t_1) - y(t_0) - (t_1 - t_0) \cdot f(t_0, y_0) = \frac{f'(c)}{2} \cdot (t_1 - t_0)^2, \quad c \in [t_0, t_1].$$

$$|y(t_1) - y(t_0) - (t_1 - t_0) \cdot f(t_0, y_0)| \leq \frac{M}{2} \cdot (t_1 - t_0)^2$$

Conclusion:

The error in one-step is proportional to Δt^2

Ex: $y' = t \cdot \sin(y)$, $y(-5) = \pi/2$, $\Delta t = 0.5$,

(written after lecture)

$$\begin{aligned} y'' &= \sin(y) + t(-\cos y) \cdot y' \\ &= \sin(y) - t \cos(y) t \sin(y) \\ &= \sin(y) - t^2 \cos(y) \sin(y) \end{aligned}$$

so

$$|y''| \leq 1 + |t^2| \leq 1 + (-5)^2 = 26 = M$$

and Error in 1-step $\leq \frac{M}{2} \cdot \Delta t^2 = \frac{26}{2} \cdot \left(\frac{1}{2}\right)^2 = \frac{13}{4} = 3.25$.

Ex: non-trivial M but easier than the other

$$y' = \underbrace{t \cos(t)}_{f(t)}$$

$$t_0 = 0, \quad t_f = 100$$

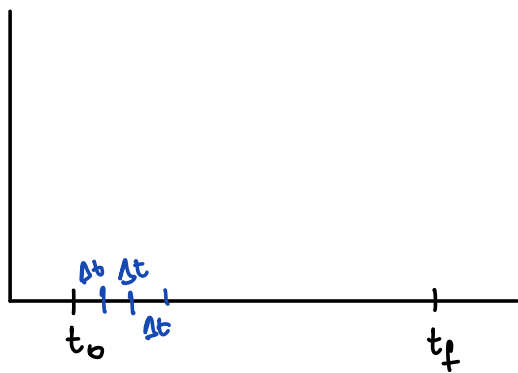
$$\text{Error} \leq 1.$$

$$\begin{aligned} |f'| &= |\cos(t) - t \sin(t)| \leq 1 + |t \sin(t)| \\ &\leq 1 + t \leq 1 + 100 \end{aligned}$$

$$t \in [0, 100]$$

$$M = 101$$

Conclusion: error from t_0 to t_f . $|f'| \leq M$



t_0 = initial time

t_f = final time

$$|y_{k+1} - y_k - \Delta t f(t_k, y_k)| \leq \text{Error}$$

$$\text{Error} \leq \frac{M}{2} (\Delta t)^2 \cdot \frac{(t_f - t_0)}{\Delta t}$$

$$\text{Error} \leq \frac{M}{2} (t_f - t_0) \cdot \Delta t$$

$$f' = y''$$

Other methods exist!

- `ode45` in MATLAB is based on the Runge-Kutta method (4,5)
- `ode113` in MATLAB has variable step-size (up to 13)

Ex. Given that $y(t) = t^2$ is the actual solution of $y' = \frac{2y}{t}$, $y(2) = 4$, which stepsize Δt guarantee error ≤ 0.001 for computing $y(7)$ using Euler's Method? (largest possible)

a) 0.02

b) 0.002

~~c) 0.0002~~

d) 0.00002

0.0002

$$\text{Error} \leq \frac{M}{2} \cdot (t_f - t_0) \cdot \Delta t$$

$$|f'| = |y''| = |(t^2)''| = |(2t)'| = 2 = M.$$

$$\frac{M}{2} \cdot (t_f - t_0) \cdot \Delta t \leq 0.001$$

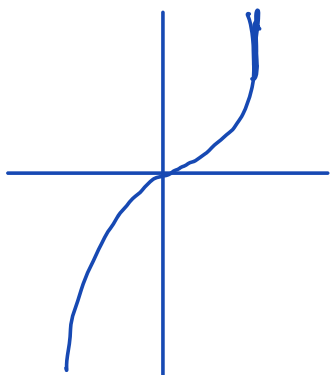
$$\Delta t \leq \frac{0.001}{\frac{7-2}{2}} = \frac{0.001}{5} = 0.0002$$

Existence and Uniqueness.

MOTIVATION:

$$a) \begin{cases} \frac{dy}{dt} = 5(1+y^2) \\ y(0) = 0 \end{cases} \quad \int_0^y \frac{dy}{1+y^2} = \int_0^t 5 dt \Rightarrow \arctan y = 5t$$

$$y = \tan(5t) \quad t \in \left(-\frac{\pi}{10}, \frac{\pi}{10}\right)$$



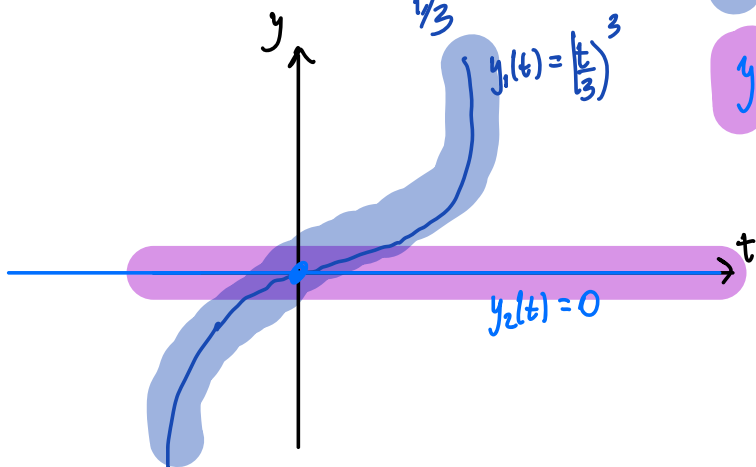
it doesn't exist for all t .

$$b) \begin{cases} y' = y^{2/3} \\ y(0) = 0 \end{cases} \quad \int_0^y y^{-2/3} dy = \int_0^t dt = t$$

$$\frac{y^{1/3}}{1/3} = t \Rightarrow y = \left(\frac{1}{3}t\right)^3$$

$$y = \left(\frac{1}{3}t\right)^3$$

$$y = 0$$



④ Uniqueness.

$$\bullet \frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

$$y_1(t), y_2(t)$$

$$y_1(t_0) = y_2(t_0)$$

if the theorem holds, then

$$y_1 = y_2$$

Theorem: Say you have

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

• if $f(t, y)$ is continuous at (t_0, y_0)

• if $\frac{\partial f}{\partial y}(t, y)$ is continuous at (t_0, y_0)

then there is a solution $\gamma: (t_0 - \varepsilon, t_0 + \varepsilon) \rightarrow \mathbb{R}$ s.t. it satisfies the initial value problem, $\varepsilon > 0$.

Ex: $\frac{dy}{dt} = \frac{1}{(y+1)(t-2)}, \quad y(0) = 0$

"Homework": ☐ CHECK CONDITIONS FOR THE THEOREM

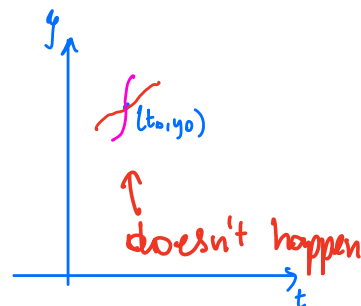
☐ FIND THE PARTICULAR SOLUTION

Curiosity:
Contraction Principle.
if $T: \mathbb{R} \rightarrow \mathbb{R}$,

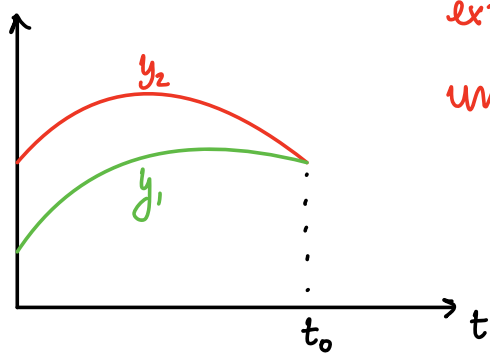
$$|T(x) - T(y)| < \lambda |x - y|$$

for $0 < \lambda < 1$,
then there is
an unique point
s.t.

$$T(\bar{x}) = \bar{x}.$$



Question: what happens if two solutions for an ODE, say y_1, y_2 , $y' = f(t, y)$, have the same value at $t = t_0$?



exist. S1. f is continuous

uniq. S2. $\frac{\partial f}{\partial y}$ is continuous

S1 \ S2	True	False
True	a	b
False	c	d

f cont.

$$\begin{cases} y' = 5/(1+y^2) \\ y(0) = 0 \end{cases}$$

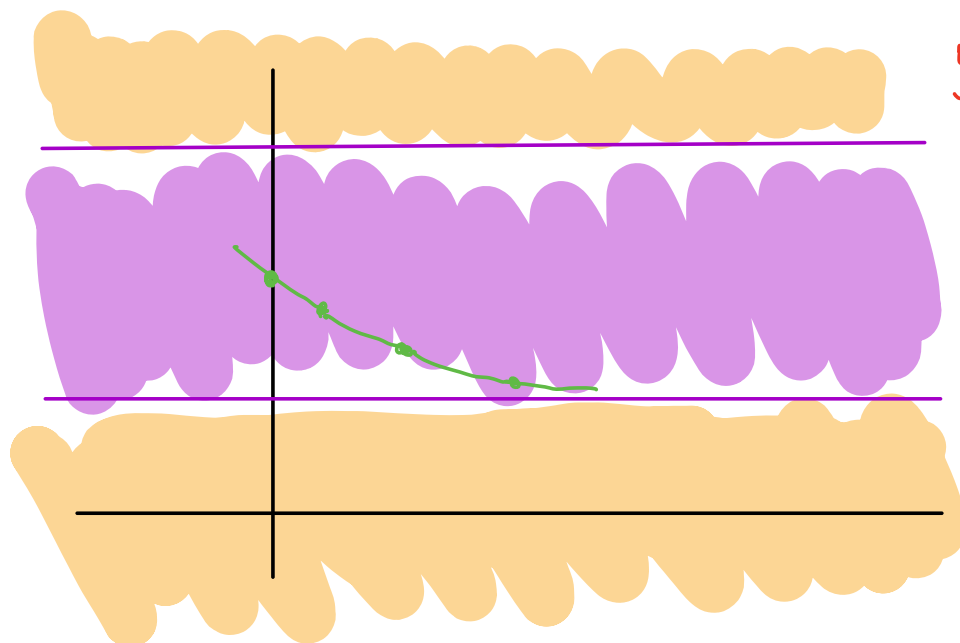
f -cont.

$$\begin{cases} y' = y^{2/3} \\ y(0) = 0 \end{cases}$$

$$f(t, y) = f(y)$$

$$\frac{\partial f}{\partial y} = \begin{cases} 10y \\ \frac{2}{3} \cdot y^{-1/3} \end{cases} \text{ NO!}$$

(weak) conclusion: equilibrium solutions divide the solutions into different regions.



$$y' = ky(1 - y/N)$$

$$= ky - \frac{k}{N} y^2$$

• is f cont? ✓

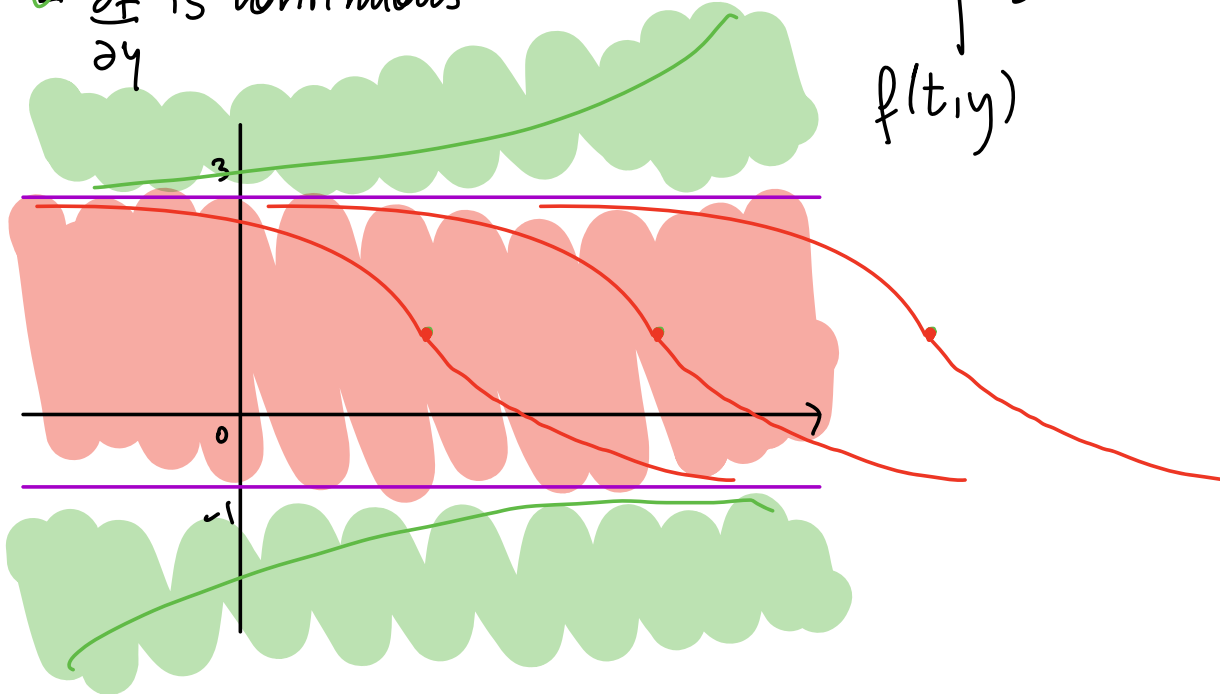
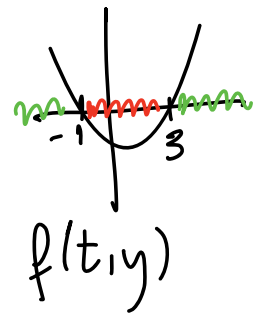
• is $\frac{\partial f}{\partial y}$ cont? ✓

Ex: equilibrium matters.

$$\frac{dy}{dt} = (y+1)(y-3)e^{\cos^2(42y)} \text{ is hard to solve but if}$$
$$f(t,y) = (y+1)(y-3)e^{\cos^2(42y)}$$

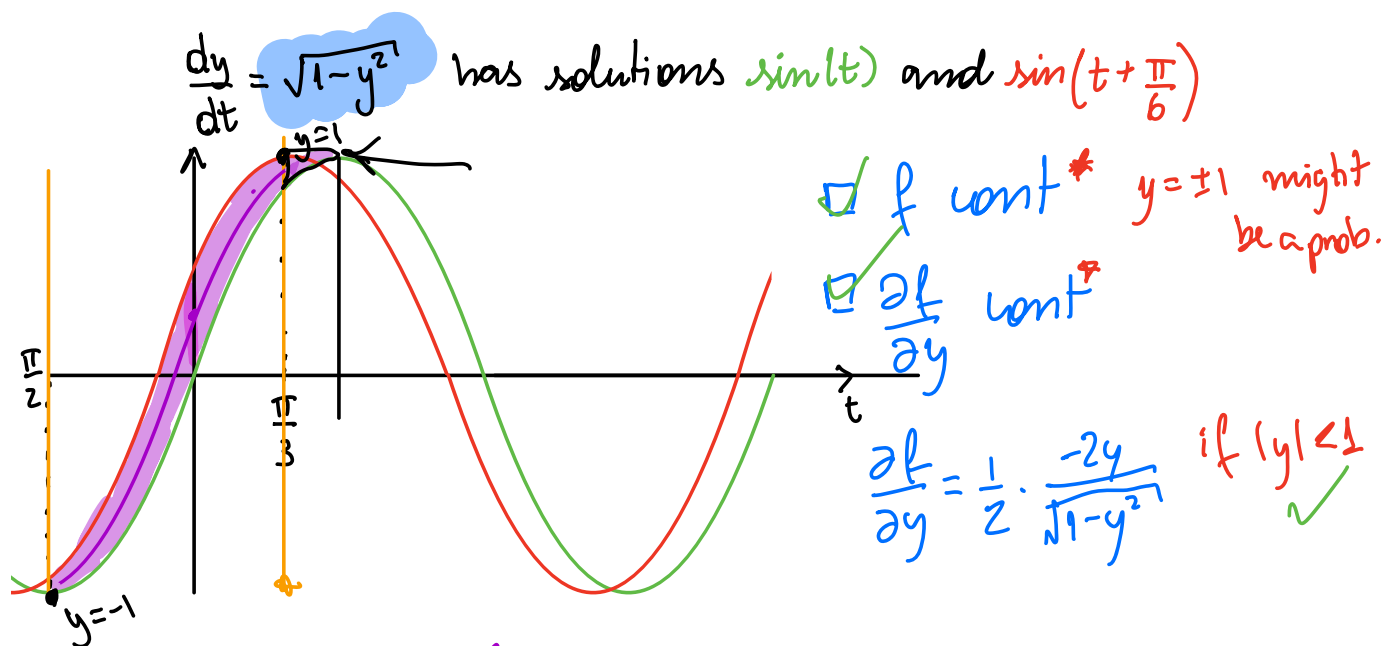
✓ f is continuous

✓ $\frac{\partial f}{\partial y}$ is continuous



Ex: Everyone matters.

$$f(x) = \sqrt{x}$$



Choose the only true statement:

~~a)~~ there is no problem here.

b) f is not continuous for $t=0, y \in [0, \frac{1}{2}]$

c) $\frac{\partial f}{\partial y}$ is not continuous for $t=0, y \in [0, \frac{1}{2}]$

d) the instructor is not funny.

Q: What can you say for solutions $y(0) \in (0, \frac{1}{2})$?

A: They are bounded below by $\sin(t)$
and bounded above by $\sin(t + \frac{\pi}{6})$
in the interval $[-\frac{\pi}{2}, \frac{\pi}{3}]$.

1.3, 1.4., 1.5. Week 1: L1+L2 ~ 1-73.

lab 1: today 23:59.

postclass 2: monday 6pm.

preclass 3: " "

Homework: 5-9 10

 Lecture 2. Lect. 1.